

ARCH 594A Research Seminar—AI for Architecture

Spring 2026

Course Description	This research seminar introduces artificial intelligence in architecture through the study of foundational theories, methods, and computational approaches. The course examines machine learning, including deep learning and physics-informed methods, graph-based learning, computer vision, and generative and agentic AI systems, with emphasis on how these approaches can be understood, evaluated, and applied in architectural design, performance modeling, and design research workflows. Students develop methodological understanding of AI-based inquiry, including data representation, model development, evaluation, and critical interpretation, while building the conceptual foundation needed to apply these techniques in research and design contexts. The seminar also explores ethical, societal, and governance questions surrounding AI in design and practice. Note: Offered concurrently with the ARCH508: AI Research Studio—AI in Performance-driven Design, the seminar forms part of an integrated learning environment linking theory and practice.
Number of Credits	3
Time and Location	Time: W 8:30 – 11:20 AM Location: ARC 110
Instructor	Narjes Abbasabadi, PhD. Assistant Professor Office location: Architecture Hall G053 E-mail address: nabbasab@uw.edu Office Hours: By appointment Zoom address: Link Office Hours: By appointment
Assistant	Hasib Hossain E-mail address: hasib97@uw.edu Office Hours: By appointment
Course Format	The course is a seminar with integrated lectures and computing lab exercises
Learning Objectives	Broad goals guiding the course design, content, and instructional approach: This seminar develops students' foundational and critical understanding of AI as a methodological framework for architectural research and design. Through theoretical instruction, case-based discussion, methodological analysis, and computational exercises, students build the capacity to understand, assess, and prepare to apply AI approaches in relation to design, performance, and human-centered questions in architecture.
Learning Outcomes	By the end of this course, students will be able to: 1. Explain core AI concepts, methods, and workflows relevant to architecture, including machine learning, deep learning, physics-informed methods, graph-based learning, computer vision, generative AI, and agentic systems.



2. Interpret and assess AI approaches in relation to performance modeling, architectural design, and research.
3. Formulate research questions that integrate AI methods with architectural domain knowledge.
4. Analyze data, model behavior, and performance metrics to evaluate the capabilities, limitations, and potential applications of AI systems.
5. Critically assess ethical, social, and governance issues in AI, including bias, data use, authorship, and responsible application.

Course Expectations:

Students are expected to:

- Prepare and engage by completing assigned readings and participating actively in discussions and activities.
- Contribute thoughtfully through informed questions and constructive engagement with peers.
- Demonstrate critical thinking in written, verbal, and analytical work, including reflection on AI methods and their implications in design.
- Engage technically with computational tools and tutorials (e.g., Python, data analysis, introductory ML workflows).
- Communicate professionally in all submissions and interactions, and meet established deadlines.
- Maintain a respectful and collaborative environment that supports peer learning and constructive dialogue.

Course Expenses

Students are not expected to incur significant additional expenses for this course. Required software and computational tools (e.g., Python, Google Colab, and other platforms) are available at no cost or through University of Washington resources. Optional expenses may include minor costs for cloud computing services or specialized tools, depending on individual project needs. These are not required and will be discussed in advance if relevant.

If you have questions or concerns about access to required resources, please contact the instructor.

Course Resources

Required readings include selected chapters from *Artificial Intelligence in Performance-Driven Design: Theories, Methods and Tools* (Abbasabadi & Ashayeri, 2024) and *Introduction to Machine Learning, 4th Edition* (Alpaydin, 2020), along with additional readings from relevant books and journal articles.

Students are not required to purchase these materials. Readings will be provided via Canvas or accessed through the University of Washington Library as needed. Specific readings are assigned regularly and detailed in each session on Canvas.

Additional course resources including the syllabus, schedule, assignments, required readings and other course materials are available through the Canvas site for this course ([Canvas site](#)).

Grading

Grades for this course are based on participation, reflections on readings and lectures, and computational assignments, and follow the University of Washington's Credit/No Credit grading system (for more information, see: https://www.washington.edu/students/gencat/front/Grading_Sys.html)

Individual assignments and activities throughout the quarter will receive numerical grades. These evaluations are used to communicate the quality of a student's work, level of engagement, and overall progress.

Percentages of final grade will be as follows:

- Participation — 30%
- Reflections (Readings & Lectures) — 30%
- Computational Assignments — 40%

Policies:

- Late work may be penalized without prior approval.
- Active participation and preparation are expected.
- Work must demonstrate clarity, rigor, and proper attribution (including AI use).
- No extra credit is offered.

See more information on incomplete grade policy here:

<https://registrar.washington.edu/grades/incomplete-grade-policy/>

Academic Integrity

The University takes academic integrity very seriously. Behaving with integrity is part of our responsibility to our shared learning community. If you're uncertain about if something is academic misconduct, ask me. I am willing to discuss questions you might have.

Acts of academic misconduct may include but are not limited to:

- Cheating (working collaboratively on quizzes/exams and discussion submissions, sharing answers, and previewing quizzes/exams)
- Plagiarism (representing the work of others as your own without giving appropriate credit to the original author(s))
- Unauthorized collaboration (working with each other on assignments)

Concerns about these or other behaviors prohibited by the Student Conduct Code will be referred for investigation and adjudication by (include information for specific campus office).

Students found to have engaged in academic misconduct may receive a zero on the assignment (or other possible outcome).

Disability Accommodation

Your experience in this class is important to me. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law. If you have already established accommodations with Disability Resources for Students (DRS), please activate your accommodations via myDRS so we can discuss how they will be implemented in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), contact DRS directly to set up an Access Plan. DRS facilitates the interactive process that establishes reasonable accommodations. Contact DRS at disability.uw.edu.

Religious Accommodation

"Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or



conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at *Religious Accommodations Policy* (<https://registrar.washington.edu/staff-faculty/religious-accommodations-policy/>). Accommodations must be requested within the first two weeks of this course using the *Religious Accommodations Request form* (<https://registrar.washington.edu/students/religious-accommodations-request/>)."

Voicing a Concern

The Department of Architecture at the University of Washington is committed to supporting all members of our community in actively addressing concerns of stigma and discrimination through a clear and accountable process. We welcome and encourage you to voice concerns as they arise and we strive to provide a safe and supportive environment during this process. Your well-being is our top priority and the Department will maintain confidentiality in all grievance matters. We are here to support you every step of the way. More information here:

<https://arch.be.uw.edu/voicing-concerns/>

Schedule

See the tentative course schedule below. As the instructor for this course, I reserve the right to adjust this schedule in any way that serves the educational needs of the students enrolled in this course. –Narjes Abbasabadi.

See [academic calendar](#) to verify the important dates/deadlines and find detailed information on them.

Last Day of Instruction: June 3, 2026

Final Examination Week: June 6–12, 2026



AI Seminar Tentative Course Schedule.

Draft April 04

WEEK	DATE	DESCRIPTION
MODULE 1 – GENERATIVE & AGENTIC AI		
Week 1	W 04/01	Introduction and Course Overview Data & Exploratory Data Analysis Guest Lecture: AI in Architectural Practice
Week 2	W 04/08	Generative AI
Week 3	W 04/14	Agentic AI
MODULE 2 – MACHINE LEARNING FUNDAMENTALS		
Week 4	W 04/22	ML, Supervised/unsupervised Learning
MODULE 3 – OTHER AI SYSTEMS		
Week 5	W 04/29	Surrogate modeling, Physics-informed ML / PINN
Week 6	W 05/06	Deep learning, Graph-based learning / GNN
Week 7	W 05/13	Deep learning /CNN, Computer Vision (CV)
Week 8	W 05/20	Reinforcement Learning
MODULE 4 – AI AND SOCIETY		
Week 9	W 05/27	Ethical, societal, and governance dimensions of AI
Week 10	W 05/03	Future of AI-driven architecture Final Review & Reflection

Note: See the course Canvas for updated and detailed schedule: [Link](#)